- Put the origin of all three arrows at the mean of your data points
- Draw PC1 and PC2 as arrows in their respective directions. Their lengths are arbitrary if you're just illustrating direction; you could draw unit-length arrows.
- Imagine a flat plane passing through the mean and spanned by PC1 and PC2.
- Project every data point onto that plane. Each point moves perpendicular to the plane onto its closest location on it.
- Ignore PC3. The small amount of variation along that direction is what you lose when reducing to two dimensions.



data:

	Frog 1	Frog 2	Frog 3	Frog 4	Frog 5	Frog 6
Gene 1	1.2	-2.6	6.5	4.2	.8	.01
Gene 2	5.4	1.1	-4.3	1.3	-6.1	4.2
Gene 3	0.2	9.5	1.5	6.7	5.5	-10.1

"rippit"

PC3 → $\lambda_1 = 5.9197$

$* \begin{pmatrix} .85249 \\ .49173 \\ .17737 \end{pmatrix}$

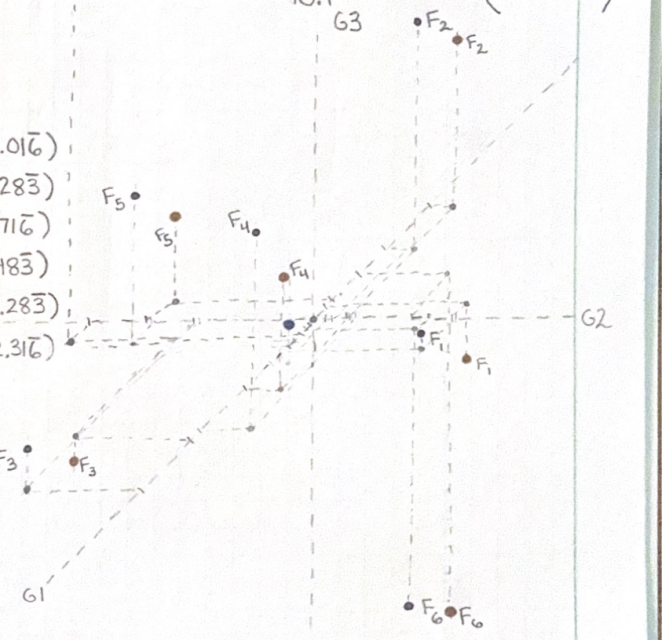
PCA w/ spectral *

1) — mean = $(1.685, .26\overline{6}, 2.21\overline{6})$

2) —
$$\begin{bmatrix} 1 & 0 & 0 & -1.685 \\ 0 & 1 & 0 & -.26\overline{6} \\ 0 & 0 & 0 & -2.21\overline{6} \\ 0 & 0 & 1 & 1 \end{bmatrix}$$
T

centered points

$F_1 \to (-.485, 5.1\overline{3}, -2.01\overline{6})$
$F_2 \to (-4.285, .83\overline{3}, 7.28\overline{3})$
$F_3 \to (4.815, -4.5\overline{6}, -.71\overline{6})$
$F_4 \to (2.515, 1.0\overline{3}, 4.48\overline{3})$
$F_5 \to (-.885, -6.3\overline{6}, 3.28\overline{3})$
$F_6 \to (-1.675, 3.9\overline{3}, -12.31\overline{6})$



3) covariance matrix

$$\begin{bmatrix} 8.62 & -4.40 & -.78 \\ -4.40 & 17.49 & -10.95 \\ -.78 & -10.95 & 40.03 \end{bmatrix}$$

4) eigen...

$$(8.62-\lambda)\begin{vmatrix} 17.49-\lambda & -10.95 \\ -10.95 & 40.03-\lambda \end{vmatrix} - (-4.40)\begin{vmatrix} -4.40 & -10.95 \\ -.78 & 40.03-\lambda \end{vmatrix} + -.78\begin{vmatrix} -4.40 & 17.49-\lambda \\ -.78 & -10.95 \end{vmatrix}$$

$$(8.62-\lambda)(\lambda^2 - 57.52\lambda + 580.23) - \Big((-4.40)(4.40\lambda - 184.673)\Big) + (-.78)(-.78\lambda + 61.82)$$

$$-\lambda^3 + 66.14\lambda^2 - 1075.75\lambda + 5000.38 + 19.36\lambda - 812.56 + .61\lambda - 48.22$$

$$-\lambda^3 + 66.14\lambda^2 - 1055.78\lambda + 4139.60$$

$\lambda_1 \approx 5.9197$, $\lambda_2 \approx 15.7113$, $\lambda_3 \approx 44.5090$

PC3 PC2 PC1

5) R3 → R2

PC1 $\lambda = 44.5090$ → $\begin{pmatrix} .0264 \\ -.3793 \\ .9249 \end{pmatrix}$

PC2 $\lambda = 15.7113$ → $\begin{pmatrix} -.5223 \\ .7837 \\ .3363 \end{pmatrix}$

preserves 91.1% of total variance